Random variables

- In any experiment, there are numerous characteristics that can be observed or measured, but in most cases an experimenter will focus on some specific aspect or aspects of a sample.
- Thus, each outcome of an experiment can be associated with a number by specifying a rule of association.
- Such a rule of association is called a ***random variable***—a variable because different numerical values are possible and random because the observed value depends on which of the possible experimental outcomes results.
- For a given sample space S of some experiment, a random variable (rv) is any rule that associates a number with each outcome in S. In mathematical language, a random variable is a function whose domain is the sample space and whose range is the set of real numbers.
- Random variables are customarily denoted by uppercase letters, such as X and Y, near the end of our alphabet.
- Lowercase letters are used to represent some particular value of the corresponding random variable. The notation $X(s) = x$ means that x is the value associated with the outcome s by the rv X.
- Any random variable whose only possible values are 0 and 1 is called a ***Bernoulli random variable***.

Two types of random variables

- A ***discrete random variable*** is a rv whose possible values either constitute a finite set or else can be listed in an infinite sequence in which there is a first element, a second element, and so on ("countably" infinite).
- A random variable is continuous if both of the following apply:
  - Its set of possible values consists either of all numbers in a single interval on the number line (possibly infinite in extent, e.g., from $-\infty$ to ∞) or all numbers in a disjoint union of such intervals (e.g., $[0, 10] \cup [20, 30]$).
  - No possible value of the variable has positive probability, that is, $P(X = c) = 0$ for any possible value c.

Probability distributions for discrete random variables

- Probabilities assigned to various outcomes in S in turn determine probabilities associated with the values of any particular rv X.
- The probability distribution of X says how the total probability of 1 is distributed among (allocated to) the various possible X values.
- **Definition**: The ***probability distribution*** or ***probability mass function (pmf)*** of a discrete rv is defined for every number x by $p(x) = P(X = x) = P(all\ s \in S : X(s) = x)$.
- In words, for every possible value x of the random variable, the pmf specifies the probability of observing that value when the experiment is performed.
- The conditions $p(x) >= 0$ and $\sum_{all\ possible\ x} p(x) = 1$ are required of any pmf.
- Example-1
  - Suppose, for example, that a business has just purchased four laser printers, and let X be the number among these that require service during the warranty period. Possible X values are then 0, 1, 2, 3, and 4. The probability distribution will tell us how the probability of 1 is subdivided among these five possible values.
    - $p(0)$ = the probability of the X value 0 = $P(X = 0)$
    - $p(1)$ = the probability of the X value 1 = $P(X = 1)$
    - and so on...
- Example-2
  - The Cal Poly Department of Statistics has a lab with six computers reserved for statistics majors.
  - Let X denote the number of these computers that are in use at a particular time of day.
  - Suppose that the probability distribution of X is as given in the following table; the first row of the table lists the possible X values and the second row gives the probability of each such value.

x	0	1	2	3	4	5	6
$p(x)$	.05	.10	.15	.25	.20	.15	.10

- The probability that at most 2 computers are in use is $P(X <= 2) = P(X = 0 \text{ or } 1 \text{ or } 2) = p(0) + p(1) + p(2) = .05 + .10 + .15 = .30$
- The probability that at least 3 computers are in use is $P(X >= 3) = 1 - P(X <= 2) = 1 - .30 = .70$

- Example-3

Six lots of components are ready to be shipped by a certain supplier. The number of defective components in each lot is as follows:

Lot	1	2	3	4	5	6
Number of defectives	0	2	0	1	2	0

One of these lots is to be randomly selected for shipment to a particular customer. Let X be the number of defectives in the selected lot. The three possible X values are 0, 1, and 2. Of the six equally likely simple events, three result in $X = 0$, one in $X = 1$, and the other two in $X = 2$. Then

$$p(0) = P(X = 0) = P(\text{lot 1 or 3 or 6 is sent}) = \frac{3}{6} = .500$$

$$p(1) = P(X = 1) = P(\text{lot 4 is sent}) = \frac{1}{6} = .167$$

$$p(2) = P(X = 2) = P(\text{lot 2 or 5 is sent}) = \frac{2}{6} = .333$$

A parameter of a probability distribution

- **Definition**: The pmf of any Bernoulli rv can be expressed in the form $p(1) = \alpha$ and $p(0) = 1 - \alpha$, where $0 < \alpha < 1$.
- Because the pmf depends on the particular value of a, we often write $p(x; a)$ rather than just $p(x)$

$$p(x; \alpha) = \begin{cases} 1 - \alpha & \text{if } x = 0 \\ \alpha & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

- **Definition**: Suppose p(x) depends on a quantity that can be assigned any one of a numbe of possible values, with each different value determining a different probability distribution. Such a quantity is called a ***parameter*** of the distribution. The collection of all probability distributions for different values of the parameter is called a ***family*** of probability distributions.
- The quantity α is a parameter.

Cumulative distribution function

- **Definition**: The **cumulative distribution function** (cdf) $F(x)$ of a discrete rv variable X with pmf $p(x)$ is defined for every number x by $F(x) = P(X <= x) = \sum_{y:y<=x} p(y)$
  For any number x, $F(x)$ is the probability that the observed value of X will be at most x.
- **Example-1**

$$p(x) = \begin{cases} .500 & x = 0 \\ .167 & x = 1 \\ .333 & x = 2 \\ 0 & \text{otherwise} \end{cases}$$

The probability that X is at most 1 is then

$$P(X \le 1) = p(0) + p(1) = .500 + .167 = .667$$

In this example, $X \le 1.5$ if and only if $X \le 1$, so

$$P(X \le 1.5) = P(X \le 1) = .667$$

Similarly,

$$P(X \le 0) = P(X = 0) = .5, \quad P(X \le .75) = .5$$

- **Example-2**

A store carries flash drives with either 1 GB, 2 GB, 4 GB, 8 GB, or 16 GB of memory. The accompanying table gives the distribution of $Y =$ the amount of memory in a purchased drive:

y	1	2	4	8	16
$p(y)$	.05	.10	.35	.40	.10

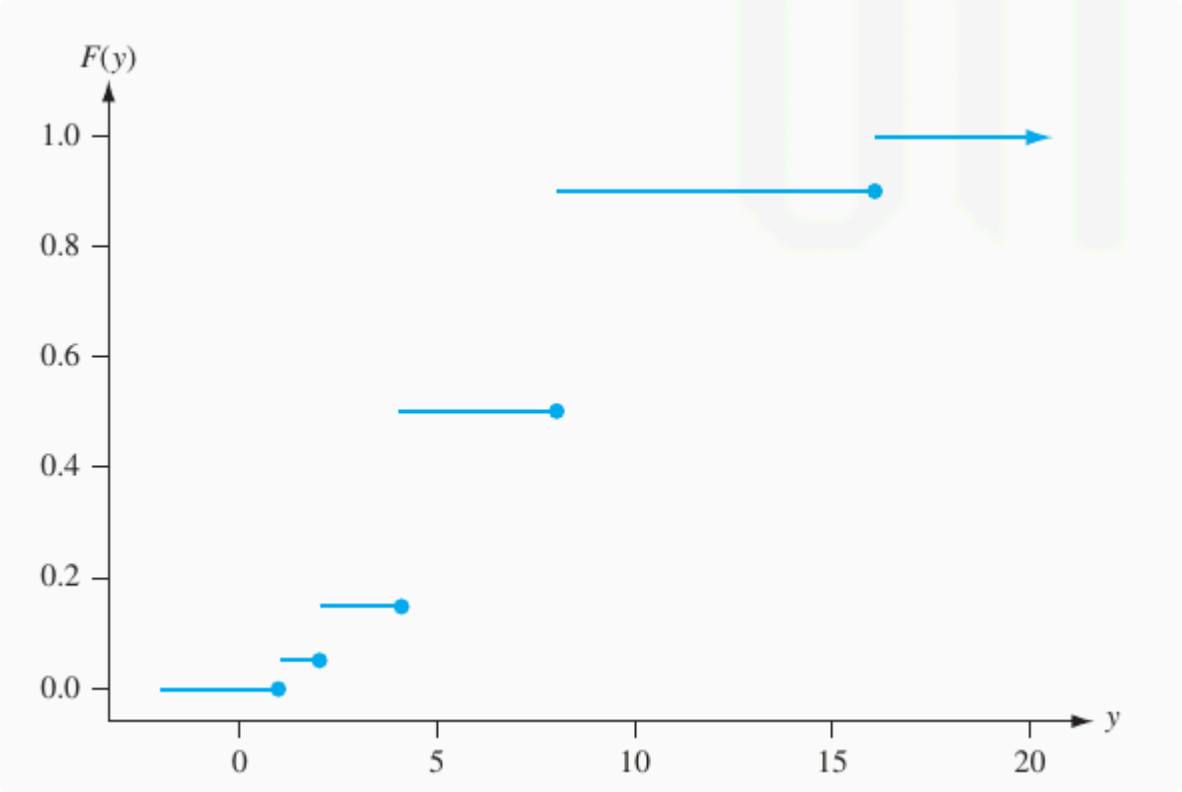
Let's first determine $F(y)$ for each of the five possible values of Y:

$$\begin{aligned} F(1) &= P(Y \le 1) = P(Y = 1) = p(1) = .05 \\ F(2) &= P(Y \le 2) = P(Y = 1 \text{ or } 2) = p(1) + p(2) = .15 \\ F(4) &= P(Y \le 4) = P(Y = 1 \text{ or } 2 \text{ or } 4) = p(1) + p(2) + p(4) = .50 \\ F(8) &= P(Y \le 8) = p(1) + p(2) + p(4) + p(8) = .90 \\ F(16) &= P(Y \le 16) = 1 \end{aligned}$$

Now for any other number y, $F(y)$ will equal the value of F at the closest possible value of Y to the left of y. For example,

$$\begin{aligned} F(2.7) &= P(Y \le 2.7) = P(Y \le 2) = F(2) = .15 \\ F(7.999) &= P(Y \le 7.999) = P(Y \le 4) = F(4) = .50 \end{aligned}$$

$$F(y) = \begin{cases} 0 & y < 1 \\ .05 & 1 \le y < 2 \\ .15 & 2 \le y < 4 \\ .50 & 4 \le y < 8 \\ .90 & 8 \le y < 16 \\ 1 & 16 \le y \end{cases}$$



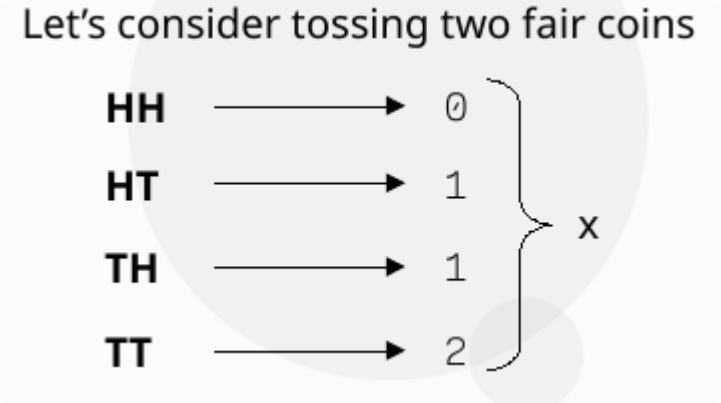
- For X a discrete rv, the graph of F(x) will have a jump at every possible value of X and will be flat between possible values. Such a graph is called a ***step function***.

The mean and the variance of a probability distribution

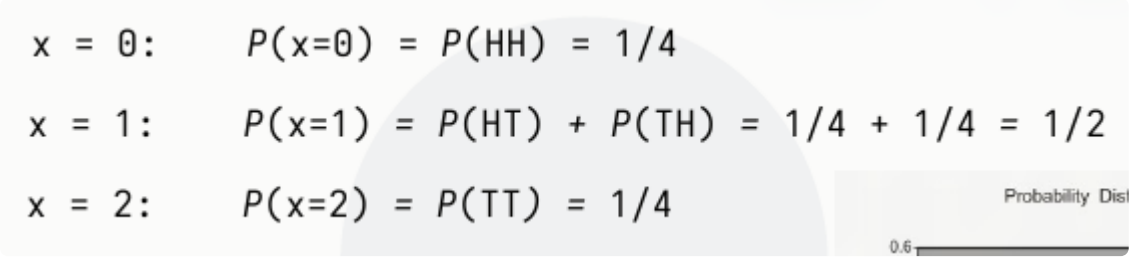
Mean

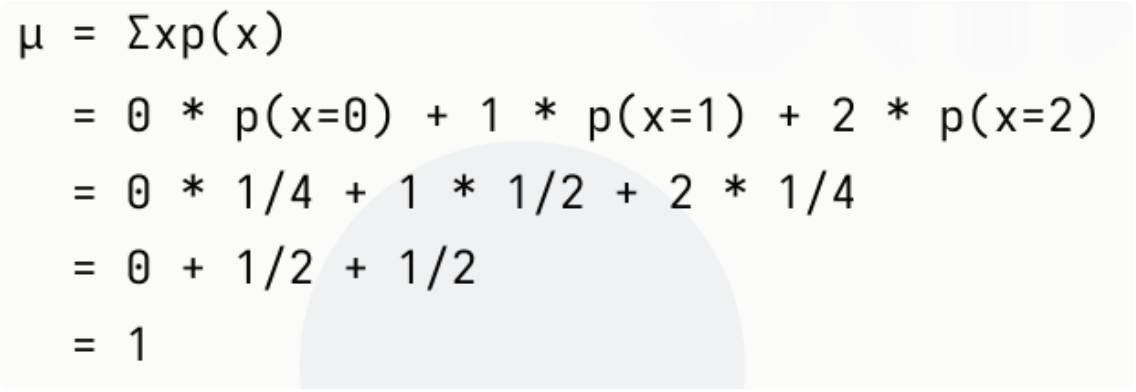
- **Definition**: Let X be a discrete rv with set of possible values D and pmf $p(x)$. The expected value or **mean value** of X, denoted by $E(X)$ or μ_X or just μ, is
  $E(X) = \mu_X = \sum_{x \in D} x * p(x)$
- μ_X doesn't necessarily need to an existing value of rv X.

- **Example-1**



Let rv X be the number of tails observed





Thus, one is most likely to get at most a single tails i.e., HT or TH when two fair coins are tossed.

Variance & Standard deviation

- **Definition**: Let X have pmf $p(x)$ and expected value μ. Then the ***variance*** of X, denoted by $V(X)$ or σ_X^2, or just σ^2 is $V(X) = \sum_D (x-\mu)^2 * p(x) = E[(X-\mu)^2]$
- The ***standard deviation*** (SD) of X is $\sigma_X = \sqrt{\sigma_X^2}$

- **Example-1**

Variance of X for Coin toss Experiment

x	(x − 1)	(x − 1)²	P(x)	(x−μ)²P(x)
0	-1	1	1/4	1/4
1	0	0	1/2	0
2	1	1	1/4	1/4

Variance, $\sigma^2 = \Sigma(x - \mu)^2 P(x) = 1/4 + 1/4 = \mathbf{0.5}$

- **Example-2**

A library has an upper limit of 6 on the number of videos that can be checked out to an individual at one time. Consider only those who check out videos, and let X denote the number of videos checked out to a randomly selected individual. The pmf of X is as follows:

x	1	2	3	4	5	6
$p(x)$	.30	.25	.15	.05	.10	.15

The expected value of X is easily seen to be $\mu = 2.85$. The variance of X is then

$$V(X) = \sigma^2 = \sum_{x=1}^{6} (x - 2.85)^2 \cdot p(x)$$
$$= (1 - 2.85)^2(.30) + (2 - 2.85)^2(.25) + \cdots + (6 - 2.85)^2(.15) = 3.2275$$

The standard deviation of X is $\sigma = \sqrt{3.2275} = 1.800$. ■

Shortcut formula for variance

- $V(X) = \sigma^2 = [\sum_D x^2 * p(x)] - \mu^2 = E(X^2) - [E(X)]^2$
- Example:

The pmf of the number X of videos checked out was given as $p(1) = .30$, $p(2) = .25$, $p(3) = .15$, $p(4) = .05$, $p(5) = .10$, and $p(6) = .15$, from which $\mu = 2.85$ and

$$E(X^2) = \sum_{x=1}^{6} x^2 \cdot p(x) = (1^2)(.30) + (2^2)(.25) + \cdots + (6^2)(.15) = 11.35$$

Thus $\sigma^2 = 11.35 - (2.85)^2 = 3.2275$ as obtained previously from the definition. ■

The Binomial Probability Distribution

Binomial experiment

- An experiment for which Conditions 1–4 are satisfied is called a ***binomial experiment***.
  1. The experiment consists of a sequence of n smaller experiments called *trials*, where n is fixed in advance of the experiment.
  2. Each trial can result in one of the same two possible outcomes (dichotomous trials), which we generically denote by success (S) and failure (F).
  3. The trials are independent, so that the outcome on any particular trial does not influence the outcome on any other trial.
  4. The probability of success $P(S)$ is constant from trial to trial; we denote this probability by p.
- **Example**: The same coin is tossed successively and independently n times. We arbitrarily use S to denote the outcome H (heads) and F to denote the outcome T (tails). Then this experiment satisfies Conditions 1–4.
- Many experiments involve a sequence of independent trials for which there are more than two possible outcomes on any one trial. A binomial experiment can then be created by dividing the possible outcomes into two groups.

The Binomial Random Variable

- **Definition**: The binomial random variable X associated with a binomial experiment consisting of n trials is defined as X $= the\ number\ of\ S's\ among\ the\ n\ trials$.
- Suppose, for example, that $n = 3$. Then there are eight possible outcomes for the experiment:
  $SSS\ SSF\ SFS\ SFF\ FSS\ FSF\ FFS\ FFF$
- From the definition of X, $X(SSF) = 2$, $X(SFF) = 1$, and so on.
- Possible values for X in an n-trial experiment are $x = 0, 1, 2, ..., n$
- We will often write $X \sim Bin(n, p)$ to indicate that X is a binomial rv based on n trials with success probability p.

The Binomial Random Variable Distribution

- Because the pmf of a binomial rv X depends on the two parameters n and p, we denote the pmf by $b(x; n, p)$.

- Consider first the case $n = 4$ for which each outcome, its probability, and corresponding x value are listed in Table 3.1. For example,

$$\begin{aligned} &P(SSFS) \\ &= P(S) * P(S) * P(F) * P(S) \ \ (independent\ trials) \\ &= p * p * (1-p) * p \ \ (constant\ P(S)) \\ &= p^3 * (1-p) \end{aligned}$$

Table 3.1 Outcomes and Probabilities for a Binomial Experiment with Four Trials

Outcome	x	Probability	Outcome	x	Probability
$SSSS$	4	p^4	$FSSS$	3	$p^3(1-p)$
$SSSF$	3	$p^3(1-p)$	$FSSF$	2	$p^2(1-p)^2$
$SSFS$	3	$p^3(1-p)$	$FSFS$	2	$p^2(1-p)^2$
$SSFF$	2	$p^2(1-p)^2$	$FSFF$	1	$p(1-p)^3$
$SFSS$	3	$p^3(1-p)$	$FFSS$	2	$p^2(1-p)^2$
$SFSF$	2	$p^2(1-p)^2$	$FFSF$	1	$p(1-p)^3$
$SFFS$	2	$p^2(1-p)^2$	$FFFS$	1	$p(1-p)^3$
$SFFF$	1	$p(1-p)^3$	$FFFF$	0	$(1-p)^4$

- In this special case, we wish $b(x; 4, p)$ for $x = 0, 1, 2, 3, 4$. For $b(3; 4, p)$ $= P(FSSS) + P(SFSS) + P(SSFS) + P(SSSF)$={number of outcomes with X=3} * {probability of any particular outcome with X=3}=$4p^3(1-p)$. Similarly, for $b(2; 4, p) = 6p^2(1-p)^2$
- Since the ordering of S's and F's is not important, the second factor in the previous equation is $p^x(1-p)^{n-x}$ (e.g., the first x trials resulting in S and the last $n-x$ resulting in F). The first factor is the number of ways of choosing x of the n trials to be S's—that is, the number of combinations of size x that can be constructed from n distinct objects (trials here).

$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x = 0, 1, 2, \ldots, n \\ 0 & \text{otherwise} \end{cases}$$

- **Example:**

Each of six randomly selected cola drinkers is given a glass containing cola S and one containing cola F. The glasses are identical in appearance except for a code on the bottom to identify the cola. Suppose there is actually no tendency among cola drinkers to prefer one cola to the other. Then $p = P(\text{a selected individual prefers } S) = .5$, so with $X =$ the number among the six who prefer S, $X \sim \text{Bin}(6,.5)$.

Thus

$$P(X = 3) = b(3; 6, .5) = \binom{6}{3}(.5)^3(.5)^3 = 20(.5)^6 = .313$$

The probability that at least three prefer S is

$$P(3 \le X) = \sum_{x=3}^{6} b(x; 6, .5) = \sum_{x=3}^{6} \binom{6}{x}(.5)^x(.5)^{6-x} = .656$$

and the probability that at most one prefers S is

$$P(X \le 1) = \sum_{x=0}^{1} b(x; 6, .5) = .109$$

The Mean and Variance of X

- For $n = 1$, the binomial distribution becomes the Bernoulli distribution.
- The mean value of a Bernoulli variable is $\mu = p$, so the expected number of S's on any single trial is p.
- Since a binomial experiment consists of n trials, intuition suggests that for $X \sim Bin(n,p)$, $E(X) = np$, the product of the number of trials and the probability of success on a single trial.
- If $X \sim Bin(n,p)$, then $E(X) = np$, $V(X) = np(1-p) = npq$, and $\sigma_X = \sqrt{npq}$ (where $q = 1 - p$).
- **Example**

If 75% of all purchases at a certain store are made with a credit card and X is the number among ten randomly selected purchases made with a credit card, then $X \sim \text{Bin}(10, .75)$. Thus $E(X) = np = (10)(.75) = 7.5$, $V(X) = npq = 10(.75)(.25) = 1.875$, and $\sigma = \sqrt{1.875} = 1.37$. Again, even though X can take on only integer values, $E(X)$ need not be an integer. If we perform a large number of independent binomial experiments, each with $n = 10$ trials and $p = .75$, then the average number of S's per experiment will be close to 7.5.

The probability that X is within 1 standard deviation of its mean value is $P(7.5 - 1.37 \le X \le 7.5 + 1.37) = P(6.13 \le X \le 8.87) = P(X = 7 \text{ or } 8) = .532$.

The Poisson Probability Distribution

- **Definition**: A discrete random variable X is said to have a ***Poisson distribution*** with parameter $\mu(\mu > 0)$ if the pmf of X is
  $p(x;\mu) = \frac{e^{-\mu} * \mu^x}{x!}$ $x = 0, 1, 2, 3, ...$
  μ -> Poisson parameter
  e -> base of the natural logarithm system; approx. 2.71828
- **Example**: Let X denote the number of creatures of a particular type captured in a trap during given time period. Suppose that X has a Poisson distribution with $\mu = 4.5$, so on average traps will contain 4.5 creatures. The probability that a trap contains exactly five creatures is

$$P(X = 5) = \frac{e^{-4.5}(4.5)^5}{5!} = .1708$$

The probability that a trap has at most five creatures is

$$P(X \leq 5) = \sum_{x=0}^{5} \frac{e^{-4.5}(4.5)^x}{x!} = e^{-4.5}\left[1 + 4.5 + \frac{(4.5)^2}{2!} + \cdots + \frac{(4.5)^5}{5!}\right] = .7029$$

Continuous Random Variables and Probability Distributions

- **Example**: If in the study of the ecology of a lake, we make depth measurements at randomly chosen locations, then X 5 the depth at such a location is a continuous rv. Here A is the minimum depth in the region being sampled, and B is the maximum depth.

Probability distributions for continuous variables

- Let X be a **continuous rv**. Then a probability distribution or probability density function (pdf) of X is a function $f(x)$ such that for any two numbers a and b with $a <= b$,
  $P(a <= X <= b) = \int_b^a f(x)dx$

- The graph of f(x) is often referred to as the ***density curve***.

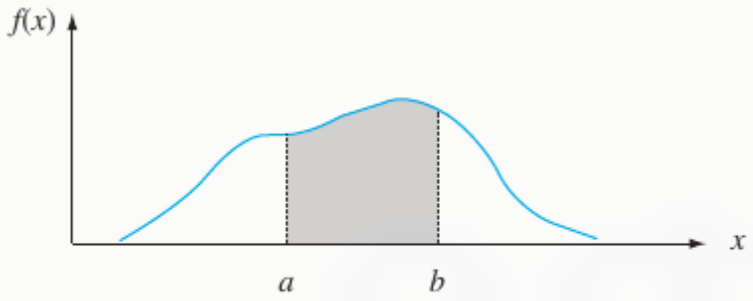


Figure 4.2 $P(a \le X \le b)$ = the area under the density curve between a and b

- For $f(x)$ to be a legitimate pdf, it must satisfy the following two conditions:
  1. $f(x) >= 0 \; \forall \; x$
  2. $\int_{-\infty}^{\infty} f(x)dx = area \; under \; the \; entire \; graph \; of \; f(x) = 1$

Uniform distribution

- A continuous rv X is said to have a ***uniform distribution*** on the interval $[A, B]$ if the pdf of X is

$$f(x; A, B) = \begin{cases} \dfrac{1}{B - A} & A \le x \le B \\ 0 & \text{otherwise} \end{cases}$$

- In the continuous case, probability density is “smeared” in a continuous fashion along the interval of possible values. When density is smeared uniformly over the interval, a uniform pdf.
- $$P(X = c) = \int_{c}^{c} f(x)dx = \lim_{\varepsilon \to 0} \int_{c-\varepsilon}^{c+\varepsilon} f(x)dx = 0$$
- The fact that $P(X = c) = 0$ when X is continuous has an important practical consequence: The probability that X lies in some interval between a and b does not depend on whether the lower limit a or the upper limit b is included in the probability calculation